

Zhaoyang Zhang

Curriculum Vitae

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📄 <https://zzyfd.github.io>

Education

2019–2024 **Ph.D. in Electronic Engineering**, *Multimedia Lab*, The Chinese University of Hong Kong.

- Under supervision of **Prof. Xiaogang Wang** and **Prof. Ping Luo**

2014–2018 **B.Sc. in Computer Science**, Wuhan University.

- Class of Outstanding Engineers (Ranking 2/62), under supervision of **Prof. Xiaohui Cui**

Research Interests

- **Efficient Reinforcement Learning**
i.e., RL-aware Distillation and RL for Image Editing.
- **Image/Video Generation and Editing**
i.e., Generative Image & Video Inpainting and Editing
- **Multi-modal and Multi-task Model**
i.e., Generalist Vision-language models
- **Efficient Deep Learning**
i.e., Network quantization, Network pruning
- **Sampled-based Dynamic Learning**
i.e., Dynamic architectures, Neural Architecture search
- **Computational Photography**
i.e., Image Denoising, Image Restoration

Working Experiences

2025.5–Now **Applied Scientist II**, Fundamental Research Team, AWS AI Lab.

- **RL-aware Distillation for LLM Reasoning.**
 - Role: First author and major contributor.
 - Submitted to ICML 2026
- **Talk2Move: Reinforcement Learning for Text-Instructed Object-Level Geometric Transformation in Scenes.**
 - Role: Corresponding author.
 - Project page: *Talk2Move*
- **Collaborators: Prof. Zhuowen Tu and Prof. Stefano Soatto**

2024.1–2025 **Senior Research Scientist, Tech Lead**, ARC Lab, Tencent.

- **Blobctrl: A Unified and Flexible Framework for Element-level Image Generation and Editing.**
 - SIGGRAPH Asia 2025, as Project Lead.
 - Project page: *BlobCtrl*
 - **VideoPainter: Any-length Video Inpainting and Editing with Plug-and-Play Context Control.**
 - SIGGRAPH 2025, as Project Lead.
 - Project page: *VideoPainter*
 - **FlexiAct: Towards Flexible Action Control in Heterogeneous Scenarios.**
 - SIGGRAPH 2025, as Project Lead.
 - Project page: *FlexiAct*
 - **Cobra: Efficient Line Art COlorization with BRoAder References**
 - SIGGRAPH 2025, as Project Lead.
 - Project page: *Cobra*
 - **BrushEdit: All-In-One Image Inpainting and Editing.**
 - TPAMI under review, as Project Lead.
 - Project page: *BrushEdit*
- 2022–2023 **Applied Scientist Intern**, *AWS Fundamental Research Team*, Amazon Web Service, Seattle, WA.
- Muskeer, a generalist Vision-Language Model with Task Explanation Prompts
 - Under supervision of **Prof. Zhuowen Tu** and **Prof. Stefano Soatto**
- 2020–2021 **Research Intern**, SenseBrain Ltd, Hong Kong.
- Efficient deep learning for computational photography. ISP optimization, e.g., HDR reconstruction, white balancing, denoising and illumination enhancement.
 - Under supervision of **Prof. Jinwei Gu**
- 2018–2019 **Research Assistant**, The Chinese University of Hong Kong, Hong Kong.
- Neural Architecture Search
 - Under supervision of **Prof. Ping Luo**

Publications

Reinforcement Learning.

- **Z Zhang**, S Jiang, Y Shen, Y Zhang, D Ram, S Yang, W Xia, Z Tu, S Soatto , “*RL-aware Distillation for LLM Reasoning.* ” , submitted to **ICML 2026**,
- J Tan, **Z Zhang**, Y Shen, J Cai, S Yang, J Wu, W Xia, Z Tu, S Soatto , “*Talk2Move: Reinforcement Learning for Text-Instructed Object-Level Geometric Transformation in Scenes* ” , submitted to **CVPR 2026**, as **Corresponding author**

Image/Video Generation - As main contributor .

- S Zhang, J Zhuang, **Z Zhang**, Y Shan, Y Tang , “*Blobctrl: A Unified and Flexible Framework for Element-level Image Generation and Editing* ” , accepted by **SIGGRAPH Asia 2025**, as **Project Lead**

- S Zhang, J Zhuang, **Z Zhang**, Y Shan, Y Tang , “*FlexiAct: Towards Flexible Action Control in Heterogeneous Scenarios* ” , accepted by **SIGGRAPH 2025**, as **Project Lead**
- J Zhuang, L Li, X Ju, **Z Zhang**, C Yuan, Y Shan , “*Cobra: Efficient Line Art Colorization with BRoAder References* ” , accepted by **SIGGRAPH 2025**, as **Project Lead**
- Y Bian, **Z Zhang**, X Ju, M Cao, L Xie, Y Shan, Q Xu, “*VideoPainter: Any-length Video Inpainting and Editing with Plug-and-Play Context Control* ” , accepted by **SIGGRAPH 2025**, as **Project Lead**
- Y Li, X Wang, **Z Zhang**, Z Wang, Z Yuan, L Xie, Y Shan, Y Zou “*Image Conductor: Precision Control for Interactive Video Synthesis* ” , accepted by **AAAI 2025**, as **Project Lead**
- Y Bian, **Z Zhang**, X Ju, M Cao, L Xie, Y Shan, Q Xu, “*NVComposer: Boosting Generative Novel View Synthesis with Multiple Sparse and Unposed Images* ” , accepted by **CVPR 2025**, as **Project Lead**

Image/Video Generation - Others .

- L Li, G Wang, **Z Zhang**, Y Li, X Li, Q Dou, J Gu, T Xue, Y Shan , “*Tooncomposer: Streamlining cartoon production with generative post-keyframing* ” , submitted to **ICLR 2026**,
- Y Li, X Li, **Z Zhang**, Y Bian, G Liu, X Li, J Xu, W Hu, Y Liu, L Li, J Cai, Y Zou, Y He, Y Shan , “*IC-Custom: Diverse Image Customization via In-Context Learning* ” , submitted to **ICLR 2026**
- X Ju*, Y Gao*, **Z Zhang***, Z Yuan, X Wang, A Zeng, Y Xiong, Q Xu, Y Shan “*MiraData: A Large-Scale Video Dataset with Long Durations and Structured Captions* ” , accepted by **NeurIPS 2024**, as **Project Lead**
- C Mou, M Cao, X Wang, **Z Zhang**, Ying Shan, Jian Zhang “*ReVideo: Remake a Video with Motion and Content Control*” , accepted by **NeurIPS 2024**
- Y Jiang*, **Z Zhang***, T Xue, J Gu “*AutoDIR: Automatic All-in-One Image Restoration with Latent Diffusion*” , accepted by **ECCV 2024**

Efficient AI - As main contributor .

- **Z Zhang**, W Shao, Y Ge, Y Shan, J Gu, X Wang, P Luo “*Cached Transformers: Improving Transformers with Differentiable Memory Cache*”, accepted by **AAAI 2024**
- **Z Zhang**, Y Jiang, W Shao, X Wang, P Luo, K Lin, J Gu, “*Real-time Controllable Denoising for Image and Video*”, accepted by **CVPR 2023**
- **Z Zhang**, Y Jiang, J Jiang, X Wang, P Luo, J Gu, “*STAR: A Semantic-aware Transformer for Real-time Image Enhancement*”, accepted by in **ICCV 2021**
- **Z Zhang**, W Shao, X Wang, P Luo “*Differentiable Dynamic Quantization with Mixed Precision and Adaptive Resolution*”, accepted by **ICML 2021 (Spotlight)**
- **Z Zhang**, J Li, W Shao, Z Peng, R Zhang, X Wang, P Luo, “*Differentiable Learning-to-group Channels via Groupable Convolutional Neural Networks*”, accepted by **ICCV 2019**

- **Z Zhang**, Z Kuang, P Luo, L Feng, W Zhang, “*Temporal Sequence Distillation: Towards Few-Frame Action Recognition in Videos*”, accepted by **ACM MM 2018**

Efficient AI - Others .

- W Shao*, M Chen*, **Z Zhang**, P Xu, L Zhao, Z Li, K Zhang, P Gao, Y Qiao, P Luo, “*OmniQuant: Omnidirectionally Calibrated Quantization for Large Language Models*”, accepted by **ICLR 2024 (Spotlight)**
- J Li, W Liu, **Z Zhang**, J Wang, T Lee, “*Model Compression for DNN-based Speaker Verification Using Weight Quantization*”, accepted by **INTERSPEECH 2023**
- W Shao, X Zhao, Y Ge, **Z Zhang**, L Yang, X Wang, Y Shan, P Luo, “*Not All Models Are Equal: Predicting Model Transferability in a Self-challenging Fisher Space*”, accepted by **ECCV 2022**
- W Shao, Y Ge, **Z Zhang**, X Xu, X Wang, Y Shan, P Luo , “*Dynamic Token Normalization Improves Vision Transformer* ”, accepted by **ICLR 2022**
- C Tao, R Lin, Q Chen, **Z Zhang**, ..., P Luo, “*FAT: Learning Low-Bitwidth Parametric Representation via Frequency-Aware Transformation*” ", accepted by **IEEE-TNNLS**
- X Yue, Z Kuang, **Z Zhang**, Z Chen, P He, Y Qiao, W Zhang, “*Boosting up Scene Text Detectors with Guided CNN*”, accepted in **BMVC 2018 (Oral)**
- W Shao, H Yu, **Z Zhang**, H Xu, Z Li, P Luo “*BWCP: Probabilistic Learning-to-Prune Channels for ConvNets via Batch Whitening*”
- W Li, **Z Zhang**, X Wang, P Luo, “*Adax: Adaptive Gradient Descent with Exponential Long Term Memory*”

Honors & Awards

- **1st place winner** (as Project Lead), GenAI Media Generation Challenge Workshop @ CVPR 2024. Tech report: <https://arxiv.org/pdf/2407.1139>
- National scholarship, Wuhan University, 2014-2016
- Outstanding intern, SenseBrain, 2019-2020
- Outstanding graduates, Wuhan University, 2018

Teaching Experiences

- 2020-2021 **Teaching Assistant**, ENGG 2740A: Differential Equations, CUHK.
- 2020-2021 **Teaching Assistant**, ENGG 2720: Complex Analysis, CUHK.
- 2019-2020 **Teaching Assistant**, ENGG 2420A: Complex Analysis and Differential Equations, CUHK.